Backup/Recovery, Snapshot creation and Restoration-33

Introduction

Backup and snapshots are essential for data protection. Backup ensures recovery in case of hardware failure or accidental deletion. Snapshots provide a quick rollback point for testing and disaster recovery.

1. Backup in Linux

Backup is the process of creating a copy of important data. There are different types of backups:

- Full Backup – Copies all files and directories.
- Incremental Backup – Copies only changes since the last backup.
- Differential Backup – Copies all changes since the last full backup.

Using tar Command

Install tar if not available

sudo yum install tar

Backup /etc directory

tar -cvpzf /backup/etc-backup.tar.gz /etc

- c → create archive
- v → verbose output
- p → preserve permissions
- z → compress using gzip
- f → specify filename

Using dd Command (Disk Image Backup)

dd if=/dev/sda of=/backup/disk.img bs=4M

- if → input file (source disk)

- of → output file (backup image)
- bs → block size (4 MB chunks for speed)

2. Recovery in Linux

Recovery is the process of restoring data from a backup.

- From tar, files can be extracted back to their original location.
- From dd, disk images can be written back to a disk.

Restore from tar Backup

tar -xvpzf /backup/etc-backup.tar.gz -C /

Restore Disk Image

dd if=/backup/disk.img of=/dev/sda bs=4M

3. Snapshot Creation (LVM)

A snapshot is a point-in-time copy of a system or volume. Unlike traditional backup, snapshots are faster and store only changes made after creation.

Features

- Quick to create and restore.
- Useful for testing and patching.
- Storage-efficient when used with LVM or virtualization platforms.

Create snapshot of logical volume

lvcreate -L1G -s -n my_snapshot /dev/vg0/my_lv

Mount snapshot

mkdir /mnt/snapshot

mount /dev/vg0/my_snapshot /mnt/snapshot

4. Snapshot Restoration (LVM)

Unmount snapshot

umount /mnt/snapshot

Merge snapshot back to original LV

lvconvert --merge /dev/vg0/my_snapshot

Activate LV

lvchange -ay /dev/vg0/my_lv

5. Backup and Snapshot in Windows Server 2019

Backup

wbadmin start backup -backupTarget:D: -include:C: -allCritical -quiet

Recovery

wbadmin start recovery -version:08/21/2025-10:00 -itemType:Volume -items:C: -recoveryTarget:C: -quiet

```
  190  susdo systemctl status tar
  191  sudo systemctl status tar
  192  sudo yum install tar
  193  tar -cvpzf /backup/etc-backup.tar.gz /etc
  194  cd
  195  tar -cvpzf /backup/etc-backup.tar.gz /etc
  196  dd if=/dev/sda of=/backup/disk.img bs=4m
  197  dd if=/dev/sda of=/backup/disk.img bs=4M
  198  cd /dev/
  199  ls
  200  dd if=/dev/disk of=/backup/disk.img bs=4M
  201  cd
  202  history
[root@localhost ~]#
```

Hyper-V Snapshot (Checkpoint)

- Open Hyper-V Manager
- Right-click VM → Checkpoint
- To restore: Right-click VM → Revert or Apply

Importance in IT Infrastructure

- Backup protects against accidental deletion, hardware failure, or ransomware.
- Recovery ensures business continuity by restoring services quickly.
- Snapshots provide developers and administrators a safe rollback option.